Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/19/2019
PI Name:	Uddin, Sardar Ph.D.		
Project Title:	Bone Remodeling under Differential Gravitational Environments		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	11794-0001	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	01/31/2019	**End Date:**	02/28/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 02/28/2022 per NSSC information. (Ed. 11/3/21). NOTE: End date changed to 2/10/2020 due to new PI Dr. David Komatsu taking over the project from Dr. Uddin; grant number remains the same. Information per F. Hernandez/ARC (Ed., 3/18/2020) NOTE: End date changed to 1/30/2022 (originally in NSSC as 2/28/2022) per F. Hernandez/ARC and actual award document (Ed., 6/12/19)		
Key Personnel Changes/Previous PI:	NOTE 3/18/2020: End date changed to 2/10/2020 due to new PI Dr. David Komatsu taking over the project from Dr. Uddin. Information per F. Hernandez/ARC.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0431		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Objective: Astronauts lose 1-1.5% of their bone mass per month during low Earth orbit missions. Spaceflight profiles of 2 years or longer, as projected for Mars missions, will significantly increase bone fragility, potentially compromising the ability of astronauts to perform their mission tasks in transit and on Mars. To facilitate human exploration of Mars, it is therefore essential to understand the underlying causes of microgravity-induced bone loss and develop effective countermeasures to sustain bone structure and mechanical integrity under microgravity conditions. Our long-term goal is to protect the skeletal health of astronauts in space environments so they can perform their mission operations and retain their quality of life upon return to Earth. The objective of the current proposal is to determine the effects of microgravity and partial microgravity on bone remodeling, as well as evaluate hypergravity as a countermeasure for microgravity-induced bone loss. Our central hypothesis is that simulated microgravity (Space 0 g) and partial microgravity (Mars 1/3 g, Moon 1/6 g) will impair bone remodeling and that hypergravity will mitigate microgravity-induced bone loss. The rationale for this proposal is the fact that microgravity and partial microgravity cause bone loss by reducing mechanical stresses on bone. Therefore, the application of periodic hypergravity will restore normal bone stresses thereby rescuing physiological remodeling. Further study will also look at recovery rate in different microgravity conditions with and without periodic hypergravity exposure. Specific Aims and Methods: The central hypothesis will be tested by completing the following two specific aims. 1-Determine the effects of microgravity and partial microgravity on bone loss and long-term recovery using rat unloading models. Our working hypothesis is that microgravity (0 g) and partial microgravity (1/3 g and 1/6 g) will alter bone remodeling leading to a long-term bone loss. Micro Computed Tomography (MicroCT) and histomorphometric analysis of bone formation and resorption will be conducted to determine microstructural changes. Mechanical testing will be used to assess structural integrity. The genomic effects of microgravity will be evaluated using RNAseq to identify differentially expressed signaling pathways. Longitudinal analysis of serum levels of bone markers will be conducted using ELISA analysis. 2-Assess the efficacy of periodic hypergravity during simulated microgravity and partial microgravity as countermeasures to mitigate microgravity-induced short term and long term effects on bone integrity. Our working hypothesis is that hypergravity will provide essential mechanical stimulation to restore microgravity and partial microgravity-induced impairment of bone remodeling. The efficacy of periodic hypergravity will be determined using bone microstructural analyses (Micro CT, bone formation, and resorption histomorphometry), mechanical analysis (three-point testing), genomic (RNA seq), and serum biomarker analysis (ELISA). Significance and Outcomes: Bone remodeling is a complex process that is tightly regulated by osteoblast-osteoclast coupling and communication. Microgravity impairs this process by increasing osteoclast associated resorption and decreasing osteoblastic bone formation. In order to continue to expand our crewed spaceflight program, it is essential to determine the effects of microgravity on bone remodeling and develop targeted countermeasures to sustain bone health in microgravity and partial microgravity. At the completion of the study, we expect to characterize the effects of microgravity and partial microgravity on bone structural and mechanical integrity, identify genomic effects of microgravity exposure, and establish the efficacy of periodic hypergravity to restore physiological bone remodeling. The outcomes of this study will fill NASA's recognized knowledge gaps (Osteo 4: We don't know the contribution of each risk factor on bone loss and recovery of bone strength, and which factors are the best targets for countermeasure application; Osteo 7: We need to identify options for mitigating early onset osteoporosis before, during and after spaceflight); and research emphasis 3 (Microbiomes of the Built Environment (MoBE) of Spacecraft Element); and 4 (Molecular and Cellular Biology Element) (Appendix G).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	With the increasing expectancy of lifespan in the US population, osteoporosis is becoming a more significant social-economical burden on American society. There is a need to develop a therapeutic solution to osteopenia and osteoporosis. The data analyses of the proposed study will identify potential therapeutic targets to restore normal bone remodeling pathways. The analysis will focus on identifying both mechanical, molecular, and combine aspects of bone remodeling and mitigation of bone remodeling. The successful completion will provide us pro-anabolic therapeutic targets to generate new bone in the osteopenic environment.
Task Progress:	The IACUC (Institutional Animal Care and Use Committee) has been approved. We have prepared 16 cages to run cohorts of 16 rats at a time. We are successfully designing the cages for a full and partial suspension of rats (1/3 and 1/6 unloading). Currently, we are working on optimizing the vests for hypergravity and partial suspension. Our expected date is to maximize the vest design by January 2020 and start the first set of the cohort by February and March. We have initiated a Principal Investigator (PI)-Self forming Biospecimens Sharing Program (BSP) to share tissues from our in vivo studies with interested collaborators. NOTE 3/18/2020: End date changed to 2/10/2020 due to new PI Dr. David Komatsu taking over the project from Dr. Uddin. See "Bone Remodeling under Differential Gravitational Environments--PI Komatsu" for subsequent reporting.
Bibliography Type:	Description: (Last Updated:)